A Wearable Device for Cataplexy Detection and Warning

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Introduction

Narcolepsy is a chronic disorder that affects the brain's ability to regulate sleep cycles. It affects roughly 25-50 in 100,000 people worldwide.[1]

Cataplexy is a sudden loss of muscle control that is a major symptom of Narcolepsy Type 1. About 20% of Narcolepsy cases involve Cataplexy.[1]

https://www.linkedin.com/pulse/competition-horizon-costly-narcolepsy-/

Problem: Dangers of Cataplexy

- **Dangers of Cataplexy:**
  - Falling asleep behind the wheel
  - Sudden collapse in a dangerous place
  - Losing muscle control while swimming

- **Narcolepsy is associated with a 1.5x increase in mortality in the US.[2]**

  - There is a need for a device that can alert individuals of an impending cataplexic episode.

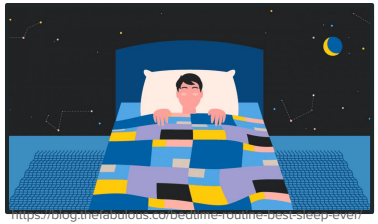
https://mydr.com.au/first-aid-self-care/narcolepsy-and-cataplexy/

Existing Solutions

- **Medication**
  - Certain medications such as Modafinil and Armodafinil are often prescribed to promote awakeness in patients.
  - Can have adverse side effects or be ineffective in preventing cataplexic attacks

https://www.freepik.com/vectors/medication

- **Lifestyle Changes**
  - Consistent bedtime, taking naps
  - Cannot guarantee improvement



https://blog.thefabulous.co/bedtime-routine-best-sleep-ever/

- **Fall Detection Devices**
  - Devices like Medical Guardian and Life Alert can call for help when someone falls from a cataplexic event
  - Cannot warn anyone BEFORE they collapse

https://www.seniorliving.org/medical-alert-systems/life-alert/reviews/

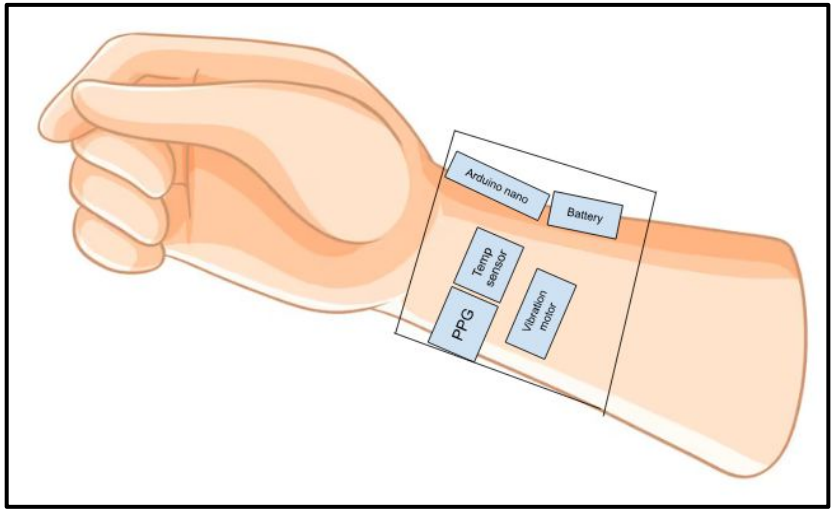
Our Approach

Through the use of a medical device, detect when cataplectic attacks happen as well as the patient's corresponding vitals to aid in preventative treatment.

<u>Physiological Reactions Before Cataplexy:</u>

Heart Rate - rapid increase 15-20% before dropping

Temperature - increases 1 C and sustains



Hardware

Photoplethysmography (PPG)

- Derive heart rate from blood volume

Thermistor*

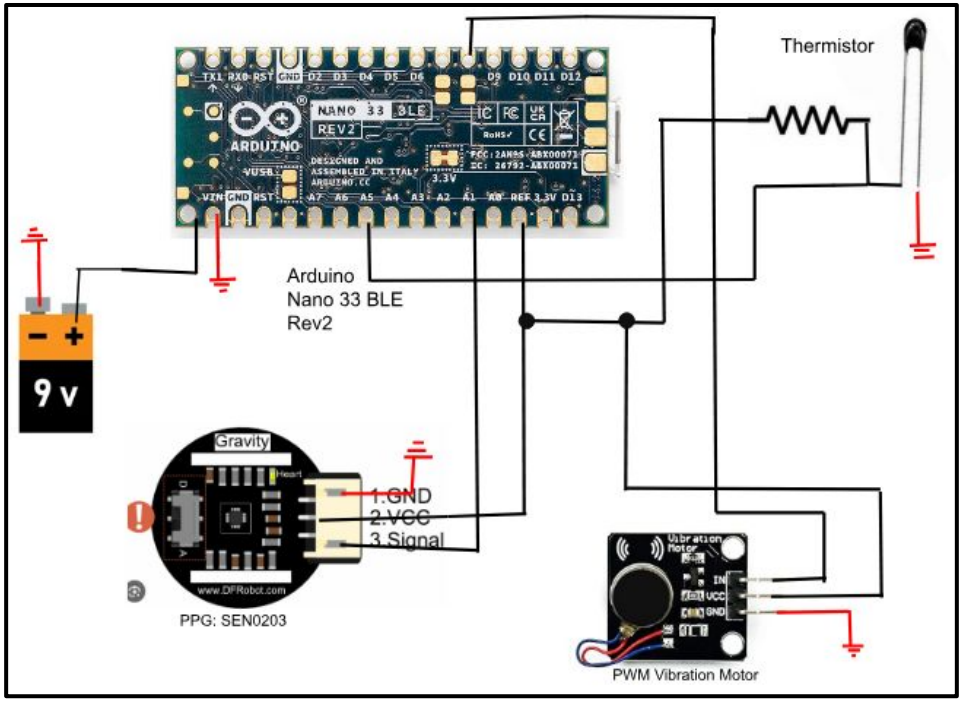
- Changes resistance based on temperature

Vibration Motor

- Haptic alert for the user

Arduino Nano

- Control and computing unit



Software

Noise Reduction

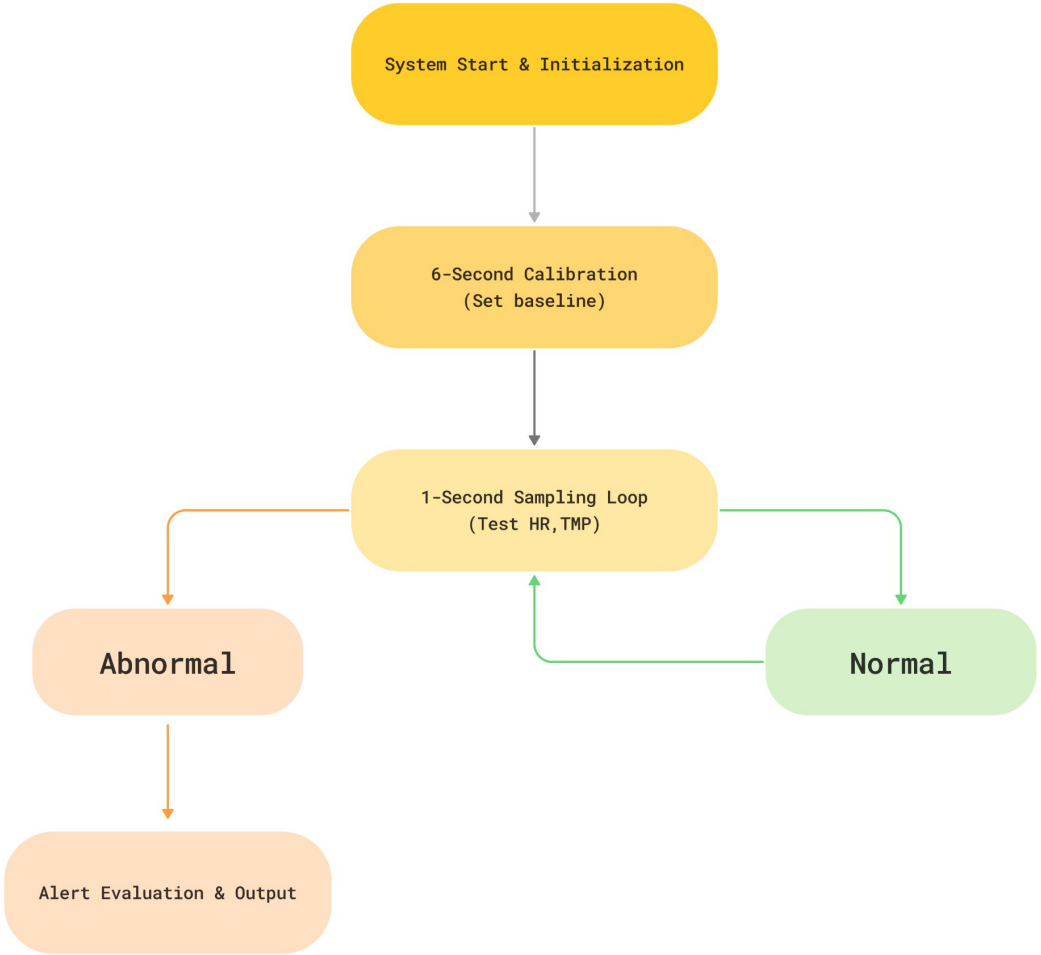
- Exponential moving average smooths heart-rate data
- Rejects unrealistic pulses to reduce jitter

Missing Data Handling

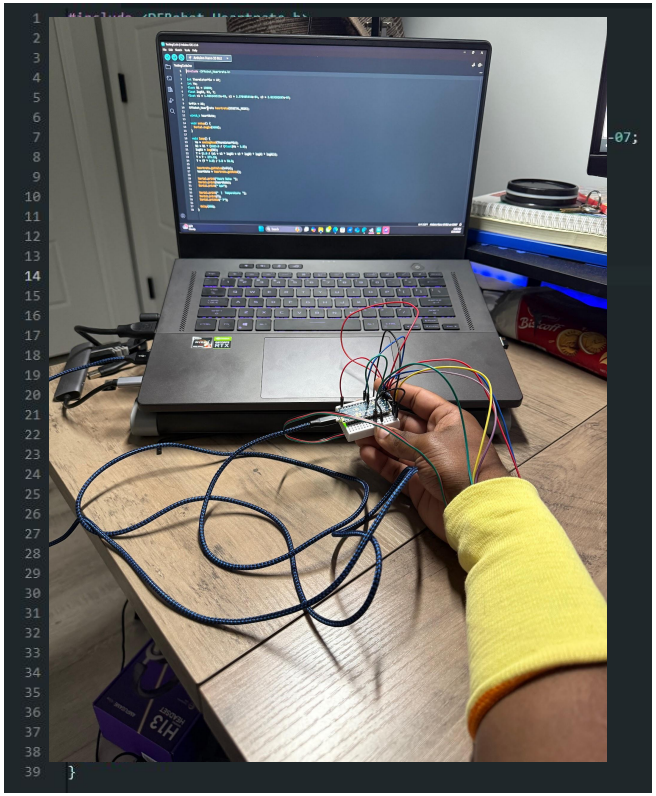
- Heart rate marked invalid if no pulse >3s
- Calibration fallback values ensure system operation

User-Independent Thresholds

- Muscle tone uses relative change, not absolute voltage
- Alerts compare readings to personalized baselines



Preliminary Testing Code



```
Heart Rate: 76 bpm  |  Temperature: 97.78 F
Heart Rate: 74 bpm  |  Temperature: 97.60 F
Heart Rate: 74 bpm  |  Temperature: 97.60 F
Heart Rate: 74 bpm  |  Temperature: 97.60 F
Heart Rate: 74 bpm  |  Temperature: 97.60 F
Heart Rate: 75 bpm  |  Temperature: 97.24 F
Heart Rate: 74 bpm  |  Temperature: 97.42 F
Heart Rate: 76 bpm  |  Temperature: 96.51 F
Heart Rate: 74 bpm  |  Temperature: 96.69 F
Heart Rate: 75 bpm  |  Temperature: 96.87 F
Heart Rate: 74 bpm  |  Temperature: 97.06 F
Heart Rate: 74 bpm  |  Temperature: 96.69 F
Heart Rate: 74 bpm  |  Temperature: 96.51 F
Heart Rate: 74 bpm  |  Temperature: 96.69 F
Heart Rate: 76 bpm  |  Temperature: 96.15 F
Heart Rate: 74 bpm  |  Temperature: 96.51 F
Heart Rate: 74 bpm  |  Temperature: 96.15 F
Heart Rate: 76 bpm  |  Temperature: 96.69 F
```

Engineering Challenges

1. EEG and EMG proved difficult due to noise and low accuracy from movement and unstable environment
2. How to secure the device to the body
   a. insert the sensors into the wristband for strong physical contact
3. Temperature sensor was not working with the Arduino Nano
   a. swapped to base Arduino thermistor (good precision in physical contact)
4. Learning Experiences
   a. I2C sensor necessity, proper soldering for MCU's, reading datasheets for component selection
5. Teamwork
   a. working around schedules, splitting up tasks, and achieving quality work in deadlines

Prototyping Steps

- Selected vitals to track and identified components
- Created wristband and schematic layout
- Developed control system and set thresholds for vitals
- Ordered components and conducted initial functional tests
- Integrated components with Arduino Nano
- Tested device accuracy with DREAMT dataset

Testing and Results

Testing Procedure

Format data from DREAMT

Run simulations in Matlab

Determine the percentage

Subject 1

TEMP	HR	Sleep_Stage
32.09	72.85	W
32.11	74.83	W
32.11	75.92	W
32.11	76.83	W
32.09	78.17	W
32.09	79.65	W
32.07	80.98	W
32.09	81.73	W
32.09	82.87	W

Subject 2

HR	TEMP	Sleep_Stage
68.52	29.59	W
68.35	29.57	W
67.92	29.43	W
67.8	29.35	W
67.4	29.15	W
66.78	29.55	W
66.35	29.57	W
65.63	29.57	W

Testing Results

```
SUBJECT 1:
  Total samples          : 2070
  Stage=R & Alert        : 480
  Stage=W & OK           : 803
  Not either (R/OK or W/Alert): 787

SUBJECT 2:
  Total samples          : 1398
  Stage=R & Alert        : 490
  Stage=W & OK           : 544
  Not either (R/OK or W/Alert): 364
```

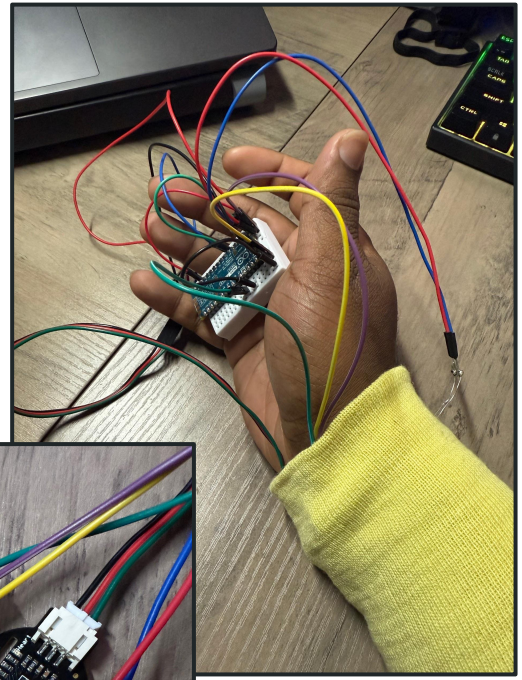
62%

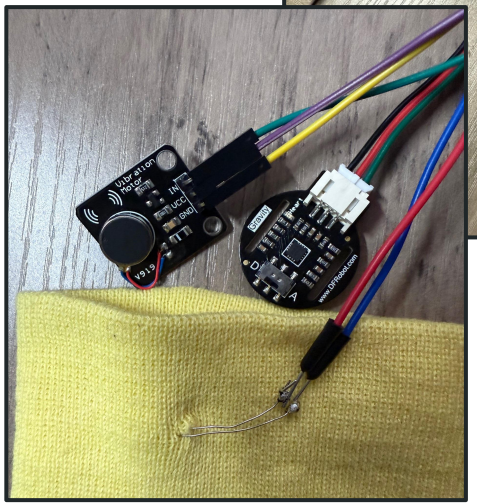
74%

Final Device

- Overall, we believe this project has good potential in consumer markets as well as research markets

- After initial research and testing, we would need to see how effectively the device performs and likely adjust our parameters once large datasets are acquired





Conclusions

Conclusions

- Cataplexy in Narcolepsy Type 1 poses severe safety risks due to sudden loss of muscle control.
- Existing treatments cannot predict events, leaving a major gap in preventative safety devices.
- The wearable wristband showed early promise by detecting jumps in heart rate and temperature.
- The device predicted relevant patterns more often than not, supporting feasibility for early warning.

Future Work

- Test with real Narcolepsy Type 1 patients to validate performance.
- Explore chest placement for improved temperature and heart-rate accuracy.
- Miniaturize the system using a PCB to improve comfort and usability.
- Refine algorithms to increase prediction consistency and reduce false alerts.

References

[1]"Narcolepsy: What it is, causes, symptoms & treatment," Cleveland Clinic, https://my.clevelandclinic.org/health/diseases/12147-narcolepsy (accessed Nov. 2, 2025).

[2] Ohayon, Maurice M et al. "Increased mortality in narcolepsy." Sleep vol. 37,3 439-44. 1 Mar. 2014, doi:10.5665/sleep.3470

[3] O. M. S. J. M. L. et al., "Prevalence and incidence of narcolepsy symptoms in the US general population," Sleep Medicine: X. [Online]. Available: https://pubmed.ncbi.nlm.nih.gov/38149177/. Accessed: Nov. 28, 2025.

[4] Cleveland Clinic, "Narcolepsy: What it is, causes, symptoms & treatment." Cleveland Clinic. [Online]. Available: https://my.clevelandclinic.org/health/diseases/12147-narcolepsy. Accessed: Nov. 28, 2025.

[5] Hackensack Sleep and Pulmonary Center, "Narcolepsy," Hackensack Sleep Center, 2025. [Online]. Available: https://hackensacksleep.com/sleep-disorders/narcolepsy/. [Accessed: Nov. 28, 2025].

[6] K. Gandhi and S. Ferdous, "Tetrad of narcolepsy type 1: Treatment and management," Cureus, vol. 16, no. 3, p. e55331, Mar. 2024, doi: 10.7759/cureus.55331.

[7] L. Barateau and Y. Dauvilliers, "Recent advances in treatment for narcolepsy," Therapeutic Advances in Neurological Disorders, vol. 12, pp. 1–12, 2019, doi: 10.1177/1756286419875622.

[8] E. Konofal, "From past to future: 50 years of pharmacological interventions to treat narcolepsy," Pharmacology Biochemistry and Behavior, vol. 241, p. 173804, 2024, doi: 10.1016/j.pbb.2024.173804.

[9] Mayo Clinic, "Narcolepsy." Mayo Clinic. [Online]. Available: https://www.mayoclinic.org/diseases conditions/narcolepsy/diagnosis-treatment/drc-20375503. Accessed: Nov. 28, 2025.

[10] H. Telangore, H. Makwana, P. Verma, M. Sharma, H. S. Mir, and U. R. Acharya, "PPG-Based Accurate Insomnia Detection System Using Convolutional Neural Networks With Self-Attention Mechanism and Gated Recurrent Units," IEEE Access, vol. 13, pp. 145474–145493, 2025, doi: https://doi.org/10.1109/access.2025.3598863.

[11] A. van der Heide et al., "Core Body and Skin Temperature in Type 1 Narcolepsy in Daily Life; Effects of Sodium Oxybate and Prediction of Sleep Attacks," Sleep, vol. 39, no. 11, pp. 1941 1949, Nov. 2016, doi: https://doi.org/10.5665/sleep.6220.